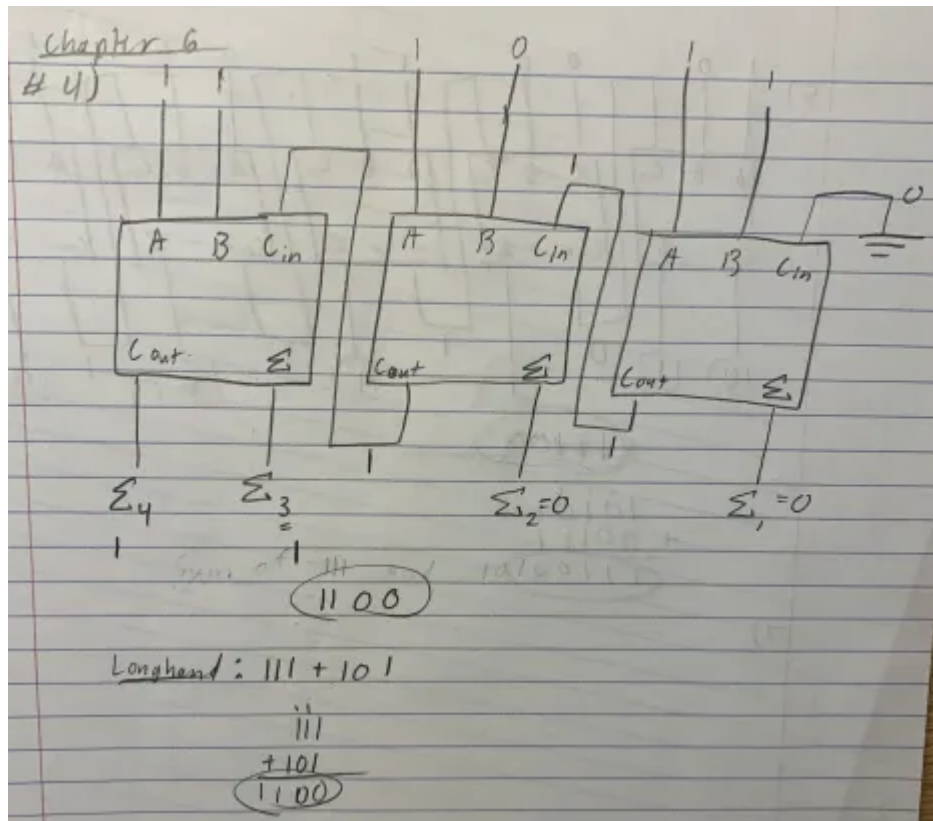


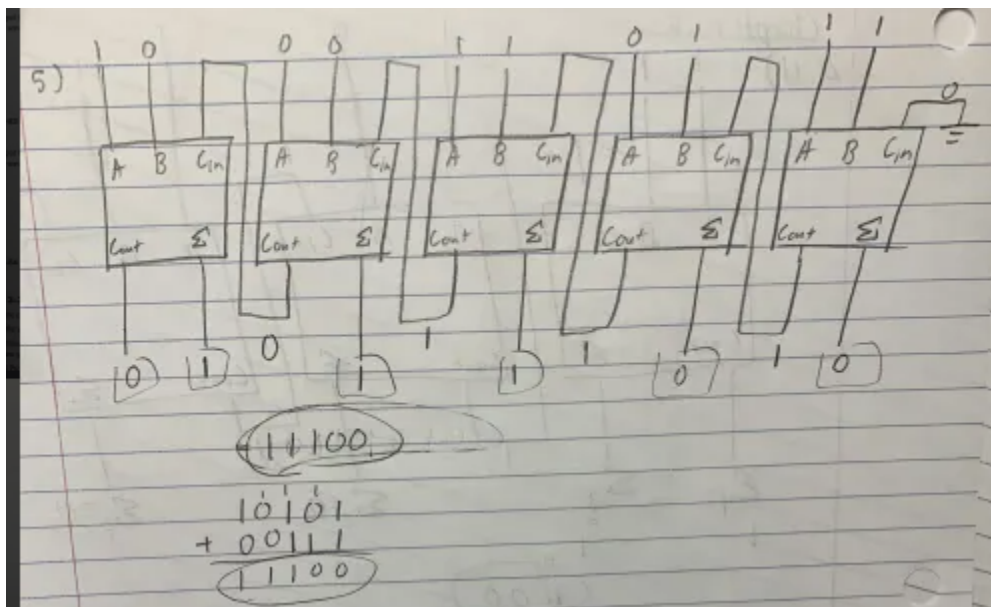
Problem Set 6

Chapter 6

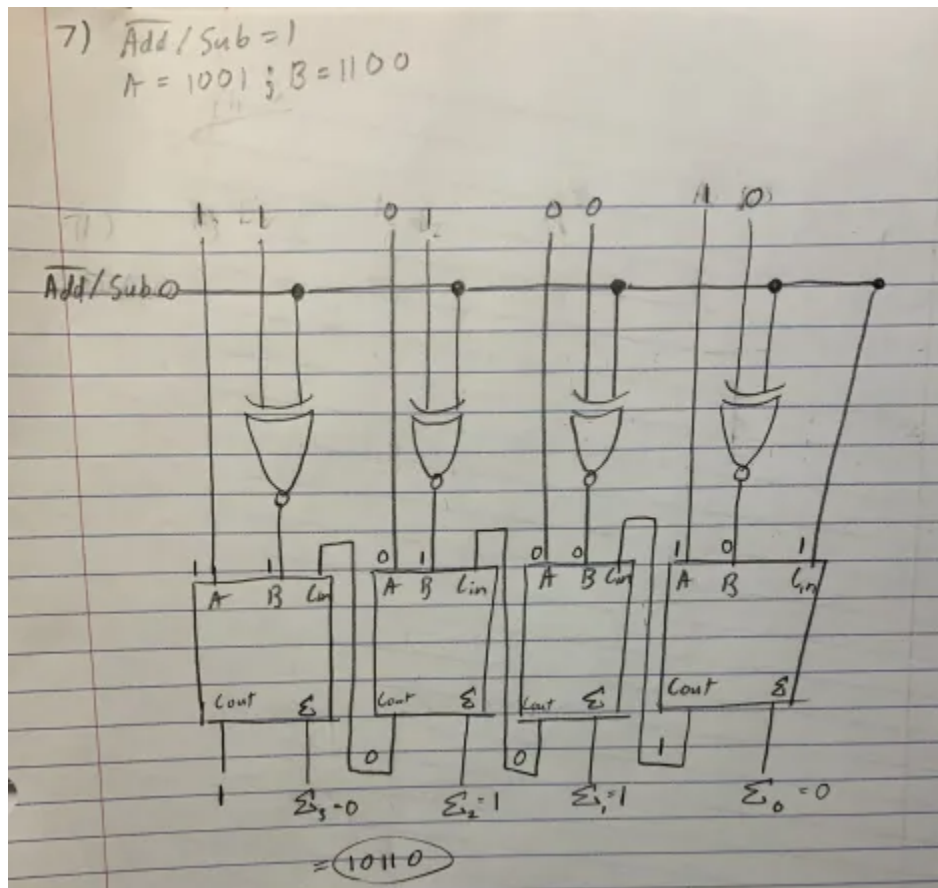
4)



5)



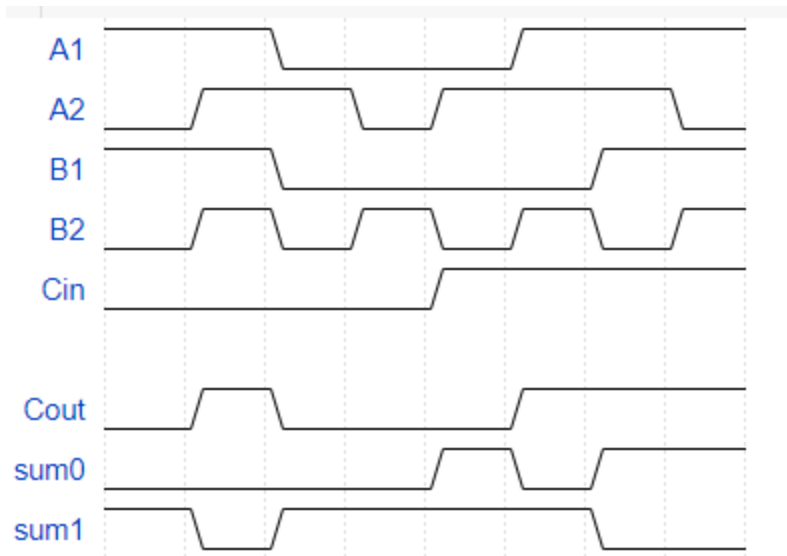
7)



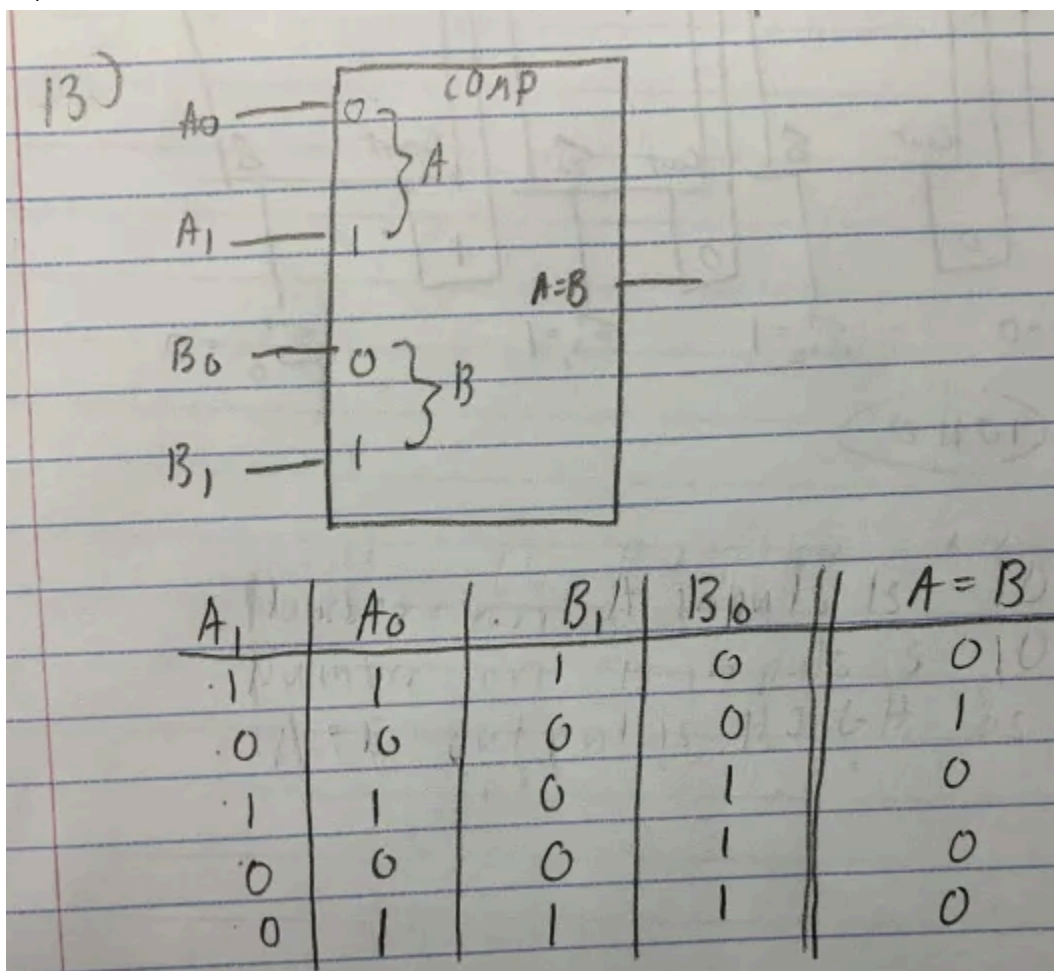
8)

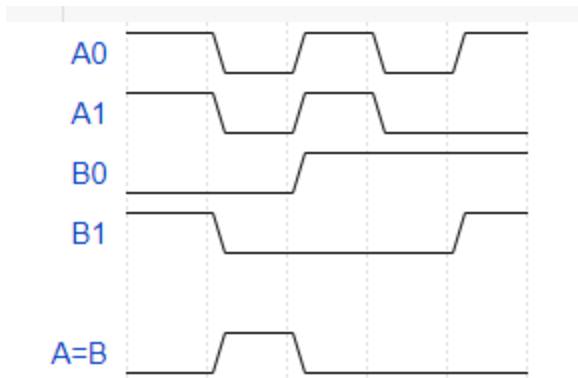
8)

A_2	A_1	B_2	B_1	C_{in}	C_{out}	Σ_1	Σ_0
0	1	0	1	0	0	1	0
1	1	1	1	0	1	0	0
1	0	0	0	0	0	1	0
0	0	1	0	0	0	1	0
1	0	0	0	1	0	1	1
1	1	1	0	1	1	1	0
1	1	0	1	1	1	0	1
0	1	1	1	1	1	0	1



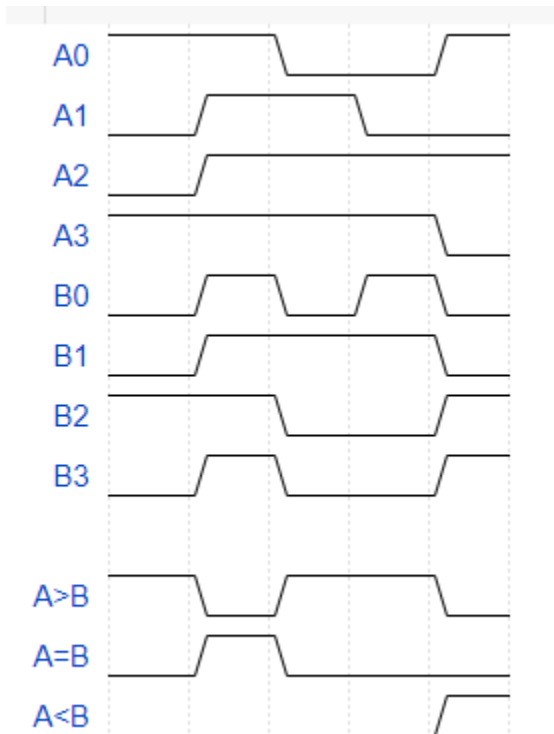
13)



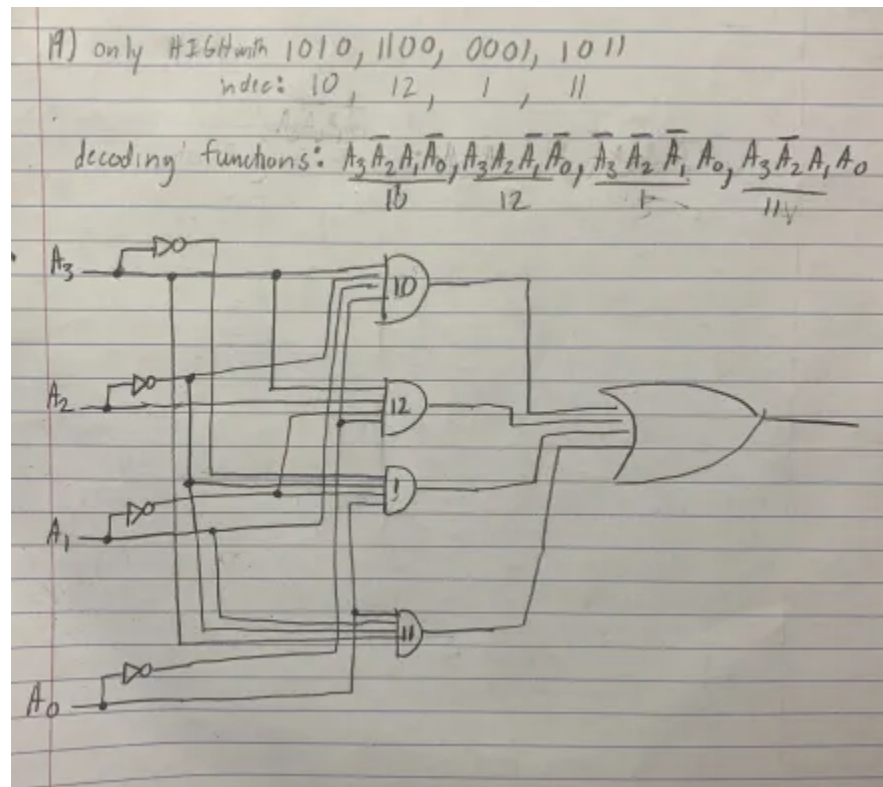


14)

14)	A ₃	A ₂	A ₁	A ₀	B ₃	B ₂	B ₁	B ₀	A > B	A = B	A < B
	1	0	0	1	0	1	0	0	1	0	0
	1	1	1	1	1	1	1	1	0	1	0
	1	1	1	0	0	0	1	0	1	0	0
	1	1	0	0	0	0	1	1	1	0	0
	0	1	0	1	1	1	0	0	0	0	1



19)



Given that there's differences between the Global and normal version of the book, I also included the problems from my book (paperback Digital Fundamentals 11th edition) just in case.

6. A CMOS technology is used to fabricate a 4-bit magnitude comparator. Which option is the most correct in the best interest of the technology used to fabricate the comparator;
 - (a) Unused input pins may be left open
 - (b) high dynamic voltage can damage the circuit
 - (c) Unused output pins may be grounded
 - (d) The input and output voltages should be between ground and V_{cc} .
7. If a 1-of-16 decoder with active-LOW outputs exhibits a LOW on the decimal 12 output, what are the inputs?
 - (a) $A_3A_2A_1A_0 = 1010$
 - (b) $A_3A_2A_1A_0 = 1110$
 - (c) $A_3A_2A_1A_0 = 1100$
 - (d) $A_3A_2A_1A_0 = 0100$
8. A BCD-to-7 segment decoder has 0100 on its inputs. The active outputs are
 - (a) a, c, f, g
 - (b) b, c, f, g
 - (c) b, c, e, f
 - (d) b, d, e, g
9. If an octal-to-binary priority encoder has its 0, 2, 5, and 6 inputs at the active level, the active-HIGH binary output is
 - (a) 110
 - (b) 010
 - (c) 101
 - (d) 000
10. MUX is a
 - (a) serial to parallel converter
 - (b) parallel to serial converter
 - (c) analog to digital converter
 - (d) digital to analog converter
11. Data selectors are basically the same as
 - (a) decoders
 - (b) demultiplexers
 - (c) multiplexers
 - (d) encoders
12. Which of the following codes exhibit even parity?
 - (a) 10011000
 - (b) 01111000
 - (c) 11111111
 - (d) 11010101
 - (e) all
 - (f) both answers (b) and (c)

PROBLEMS

Answers to odd-numbered problems are at the end of the book.

Section 6-1 Half and Full Adders

1. For the full-adder of Figure 6-4, determine the logic state (1 or 0) at each gate output for the following inputs:
 - (a) $A = 1, B = 1, C_{in} = 1$
 - (b) $A = 0, B = 1, C_{in} = 1$
 - (c) $A = 0, B = 1, C_{in} = 0$
2. What are the full-adder inputs that will produce each of the following outputs:
 - (a) $\Sigma = 0, C_{out} = 0$
 - (b) $\Sigma = 1, C_{out} = 0$
 - (c) $\Sigma = 1, C_{out} = 1$
 - (d) $\Sigma = 0, C_{out} = 1$
3. Determine the outputs of a full-adder for each of the following inputs:
 - (a) $A = 1, B = 0, C_{in} = 0$
 - (b) $A = 0, B = 0, C_{in} = 1$
 - (c) $A = 0, B = 1, C_{in} = 1$
 - (d) $A = 1, B = 1, C_{in} = 1$

Section 6-2 Parallel Binary Adders

4. For the parallel adder in Figure 6-69, determine the complete sum by analysis of the logical operation of the circuit. Verify your result by longhand addition of the two input numbers.

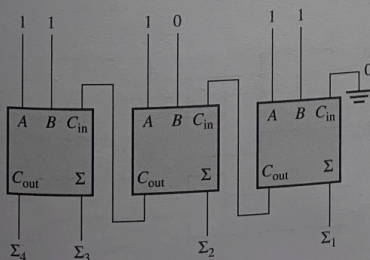


FIGURE 6-69

5. Repeat Problem 4 for the circuit and input conditions in Figure 6-70.

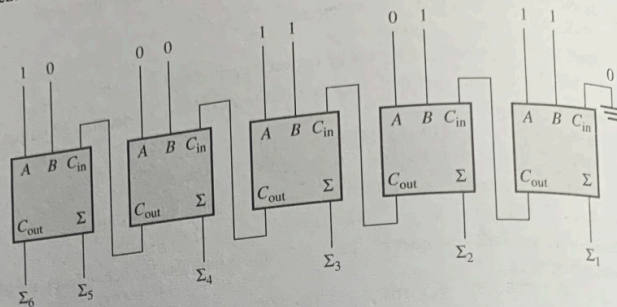


FIGURE 6-70

6. The circuit shown in Figure 6-71 is a 4-bit circuit that can add or subtract numbers in a form used in computers (positive numbers in true form; negative numbers in complement form). (a) Explain what happens when the *Add/Subt.* input is HIGH. (b) What happens when *Add/Subt.* is LOW?

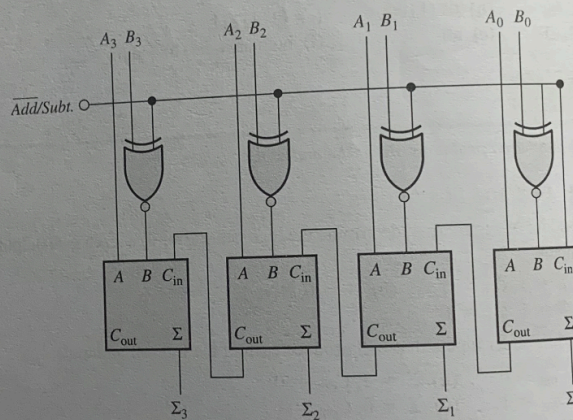


FIGURE 6-71

7. For the circuit in Figure 6-71, assume the inputs are $\overline{\text{Add/Subt.}} = 1$, $A = 1001$ and $B = 1100$. What is the output?
8. The input waveforms in Figure 6-72 are applied to a 2-bit adder. Determine the waveforms for the sum and the output carry in relation to the inputs by constructing a timing diagram.

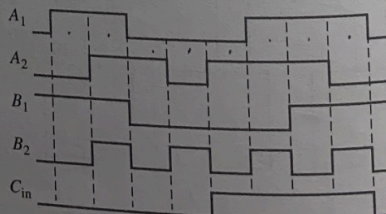


FIGURE 6-72

9. The following adder. Determine

A_1
 A_2
 A_3
 A_4
 B_1
 B_2
 B_3
 B_4

10. In the process on its pins: 11-HIGH,

Section 6-

11. Each of the gates has a propagation delay of 10 ns. Determine the output of the circuit for the following inputs:

A to Σ
 B to Σ

C

Determine

12. Show the output of the circuit for the following inputs: Figure 6-1

Section 6-

13. The waveform for the output of the circuit is $(A = B)$ w

FIGURE 6-73

14. For the 4-bit adder, the outputs are

A

A

A

A

B

B

B

B

FIGURE 6-74

9. The following sequences of bits (right-most bit first) appear on the inputs to a 4-bit parallel adder. Determine the resulting sequence of bits on each sum output.

A_1	1001
A_2	1110
A_3	0000
A_4	1011
B_1	1111
B_2	1100
B_3	1010
B_4	0010

10. In the process of checking a 74HC283 4-bit parallel adder, the following logic levels are observed on its pins: 1-HIGH, 2-HIGH, 3-HIGH, 4-HIGH, 5-LOW, 6-LOW, 7-LOW, 9-HIGH, 10-LOW, 11-HIGH, 12-LOW, 13-HIGH, 14-HIGH, and 15-HIGH. Determine if the IC is functioning properly.

Section 6-3 Ripple Carry and Look-Ahead Carry Adders

11. Each of the eight full-adders in an 8-bit parallel ripple carry adder exhibits the following propagation delays:

A to Σ and C_{out} :	40 ns
B to Σ and C_{out} :	40 ns
C_{in} to Σ :	35 ns
C_{in} to C_{out} :	25 ns

Determine the maximum total time for the addition of two 8-bit numbers.

12. Show the additional logic circuitry necessary to make the 4-bit look-ahead carry adder in Figure 6-17 into a 5-bit adder.

Section 6-4 Comparators

13. The waveforms in Figure 6-73 are applied to the comparator as shown. Determine the output ($A = B$) waveform.

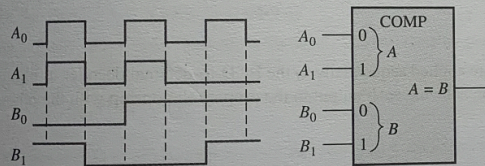


FIGURE 6-73

14. For the 4-bit comparator in Figure 6-74, plot each output waveform for the inputs shown. The outputs are active-HIGH.

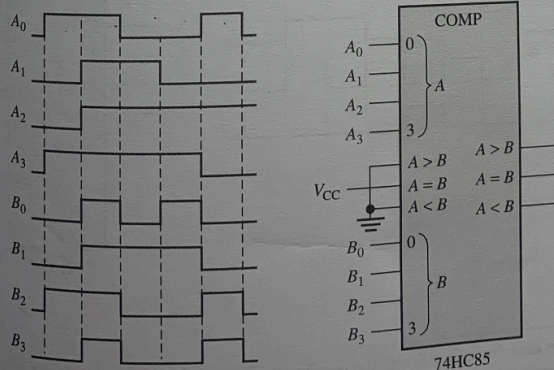


FIGURE 6-74

15. For each set of binary numbers, determine the output states for the comparator of Figure 6-21.
- (a) $A_3A_2A_1A_0 = 1100$
 $B_3B_2B_1B_0 = 1001$
- (b) $A_3A_2A_1A_0 = 1000$
 $B_3B_2B_1B_0 = 1011$
- (c) $A_3A_2A_1A_0 = 0100$
 $B_3B_2B_1B_0 = 0100$

Section 6-5 Decoders

16. Implement the following function using a 3 line to 8 line decoder.
 $S(A, B, C) = \Sigma m(1, 2, 4, 7)$ and $C(A, B, C) = \Sigma m(3, 5, 6, 7)$
 Where, S stands for 'Sum' and C stands for 'Carry In'.
17. Show the decoding logic for each of the following codes if an active-HIGH (1) output is required:
- (a) 1101 (b) 1000 (c) 11011 (d) 11100
 (e) 101010 (f) 111110 (g) 000101 (h) 1110110
18. Solve Problem 17, given that an active-LOW (0) output is required.
19. You wish to detect only the presence of the codes 1010, 1100, 0001, and 1011. An active-HIGH output is required to indicate their presence. Develop the minimum decoding logic with a single output that will indicate when any one of these codes is on the inputs. For any other code, the output must be LOW.
20. If the input waveforms are applied to the decoding logic as indicated in Figure 6-75, sketch the output waveform in proper relation to the inputs.

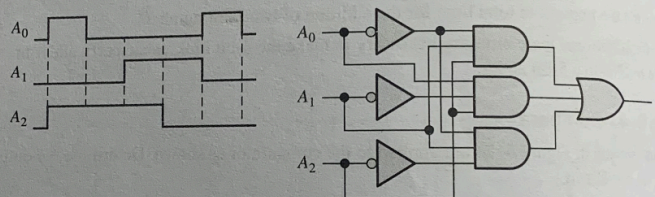


FIGURE 6-75

21. BCD numbers are applied sequentially to the BCD-to-decimal decoder in Figure 6-76. Draw a timing diagram, showing each output in the proper relationship with the others and with the inputs.

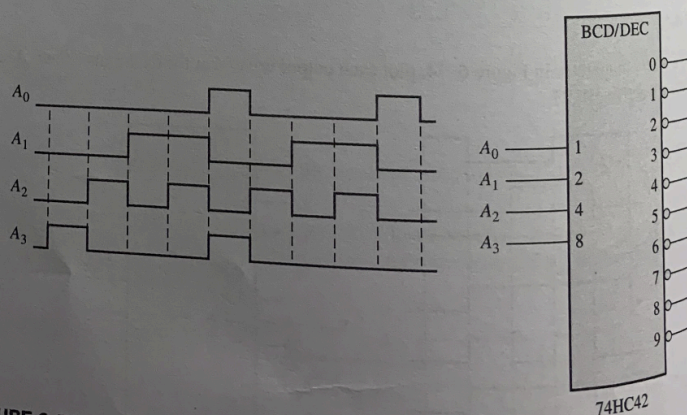


FIGURE 6-76

22. A 7-segment decoder/driver drives the display in Figure 6-77. If the waveforms are applied as indicated, determine the sequence of digits that appears on the display.